

Semantic Drift Discovery Over Time: Tracing Linguistic Evolution via Diachronic Word Embeddings and Sentential Constellations

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Abstract—Language is continually evolving, with words shifting meanings in response to technological, cultural, and societal developments. Previous computational approaches to semantic drift have largely focused on estimating the degree of semantic change for individual, isolated words. However, human communication relies heavily on composite phrases and sentences, where the collective contextual shift often dictates the trajectory of meaning evolution. In this paper, we present a comprehensive framework for tracking and visualizing diachronic semantic drift using historical word embeddings from the Google N-gram corpus (1800-1990). We introduce a novel analytic method: the "Semantic Drift Constellation," alongside a "Counterfactual Contribution Scoring" mechanism that successfully adapts the Top-K discovery paradigm to the sentential level. Our approach automatically identifies the core lexical drivers causing the overall meaning of an idiomatic expression or sentence to drift, mapping these changes across categorical semantic domains. An interactive, web-based dashboard serves as a tangible manifestation of these methodologies. Our findings extend the field of historical and computational linguistics by enabling the holistic visualization of semantic restructuring over time.

Index Terms—Natural Language Processing, Semantic Drift, Diachronic Linguistics, Word Embeddings, Data Visualization

I. INTRODUCTION

Language is not static; it is a continuously evolving system reflecting the transformations within society. Semantic drift, also referred to as semantic change, marks the evolution of word usage and meaning over time. For example, the term "broadcast" originated in agriculture to describe the scattering of seeds over a wide area, but predominantly transitioned into the technological domain during the advent of radio and modern mass media. Understanding these shifts is crucial for historical linguistics, digital humanities, and improving temporally-aware natural language processing (NLP) models.

Recent advancements in predictive language modeling and distributional semantics have facilitated large-scale empirical studies of lexical semantic change. By training word representations (embeddings) on corpora from different time periods, researchers can track a word's continuous trajectory through high-dimensional vector space. However, existing literature predominantly focuses on the trajectory of isolated words decoupled from their practical linguistic environments.

In this work, we advance the paradigm of semantic drift analysis by elevating it from individual words to complete

phrases and sentences. We introduce a robust system combining Orthogonal Procrustes alignment on historical vocabulary sets with a novel Sentential Top-K Discovery methodology.

II. RELATED WORK

The computational study of semantic change using distributed representations gained significant traction following the popularization of Word2Vec architectures. Kim et al. (2014) [1] were among the first to introduce continuous representations for tracking nuanced meaning changes dynamically. To allow the direct mathematical comparison of vectors trained independently on disparate historical corpora, Hamilton et al. (2016) [2] established the Orthogonal Procrustes method, which effectively aligns different temporal spaces into a unified coordinate system without requiring predefined bilingual dictionaries.

More recently, the paradigm has shifted from "Detection" to "Discovery". As opposed to simply measuring the degree of change for a pre-selected target word, Lee et al. (2025) [3] proposed evaluating Top-K discovery frameworks to automatically find which words shifted their contextual representations most significantly across an entire vocabulary index. Our project builds directly upon this foundational finding by extending the Top-K discovery approach into sentential boundaries, intelligently analyzing which constituent words dominate the semantic shift within a localized contextual phrase.

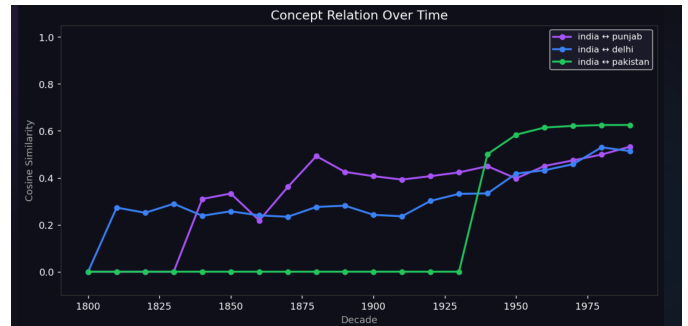


Fig. 1. Correlation between Two Words

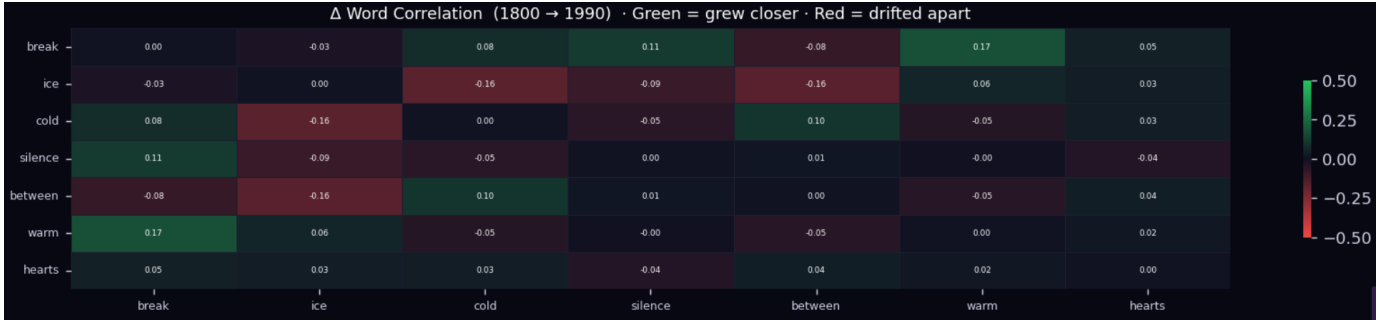


Fig. 2. Word Correlation in a sentence

III. METHODOLOGY

A. Dataset and Embeddings

We utilized the Stanford Historical Word Vectors derived from the massive Google N-gram corpus, encompassing decades from the 1800s to the 1990s. The dataset provides dense Singular Value Decomposition (SVD) and Skip-Gram representations. We opted to employ the Skip-Gram with Negative Sampling (SGNS) embeddings due to their documented superior performance in relational semantic evaluations and historical stability.

B. Temporal Space Alignment

Embeddings generated from separate corpora suffer from stochastic initializations, resulting in inherently unaligned vector spaces. Let W_t represent the matrix of word vectors at time t . We systematically align W_t to a common base time W_{base} using an orthogonal transformation matrix Q that minimizes the Frobenius norm $\|W_t Q - W_{base}\|_F$. The transformation resolves exactly through the Orthogonal Procrustes problem using standard Singular Value Decomposition.

IV. SYSTEM IMPLEMENTATION

We developed an interactive diachronic analysis dashboard designed specifically for researchers, digital anthropologists, and linguists. The architecture utilizes a Python-based backend powered by the Flask framework. To mitigate the severe memory overhead of concurrently loading multiple high-dimensional vocabularies representing distinct decades, the engine mounts embedding matrices natively to the disk using mmap.

The interactive interface features multi-tabular capabilities including:

- **Word Explorer:** Monitoring specific vocabulary drift metrics, Jaccard overlap stability, and emerging contextual neighbors.
- **Correlation Heatmaps:** Visualizing semantic proximity dynamically between a user-defined set of correlated words to trace decaying associative bonds.
- **Ranking Engine:** Iteratively uncovering the most volatile and top-drifted terminology comparing arbitrary periods.

V. INNOVATION: SENTENTIAL CONSTELLATIONS

The primary scientific contribution of our research is the implementation of the "Semantic Drift Constellation," successfully bridging independent word evolution mapping with complete localized sentence alignment.

A. Counterfactual Contribution Scoring

When evaluating the semantic evolution of an entire sentence, the standard procedure is frequently restricted to merely aggregating constituent word representations. However, this bulk aggregation obscures the true driving forces dictating the directional change. We deploy a *Counterfactual Contribution Score* calculated uniquely for every constituent word w in an arbitrary sequence S .

First, we compute the cosine distance of the sentence aggregated representation between era A and B , denoting total shift $D(S)$. We then retroactively compute the distance excluding the target word representation, $D(S_{-w})$. The final contribution score is evaluated as $D(S) - D(S_{-w})$. A high positive contribution mathematically signifies the word heavily biased the overall sentential semantic shift, allowing for algorithmic Top-K driver discovery within a phrase.

B. Domain Transition Analytics

To structurally enhance qualitative interpretability, we developed an automated domain mapping algorithm utilizing nearest-neighbor lexical overlap protocols. A taxonomy of categorized domains (e.g., Technology, Agriculture, Biology, Maritime, Medicine) acts as seed constants. A word's trailing and leading contiguous semantics are scored against these taxonomies, granting our algorithmic engine the capability to generate structural, natural-language reasoning such as "Migrated from Agriculture to Media-Communication."

VI. CONCLUSION

This research successfully constructs and implements a robust, interactive platform designed to systematically monitor semantic change spanning nearly two centuries. By introducing sentence-level Counterfactual Contribution Scoring alongside explicit Domain Transition analytics, we explicitly shifted the historical analysis paradigm from simply tracing isolated word lists to exploring the cascading inter-relational effects of semantic drift on unified meaning. Future extensions will

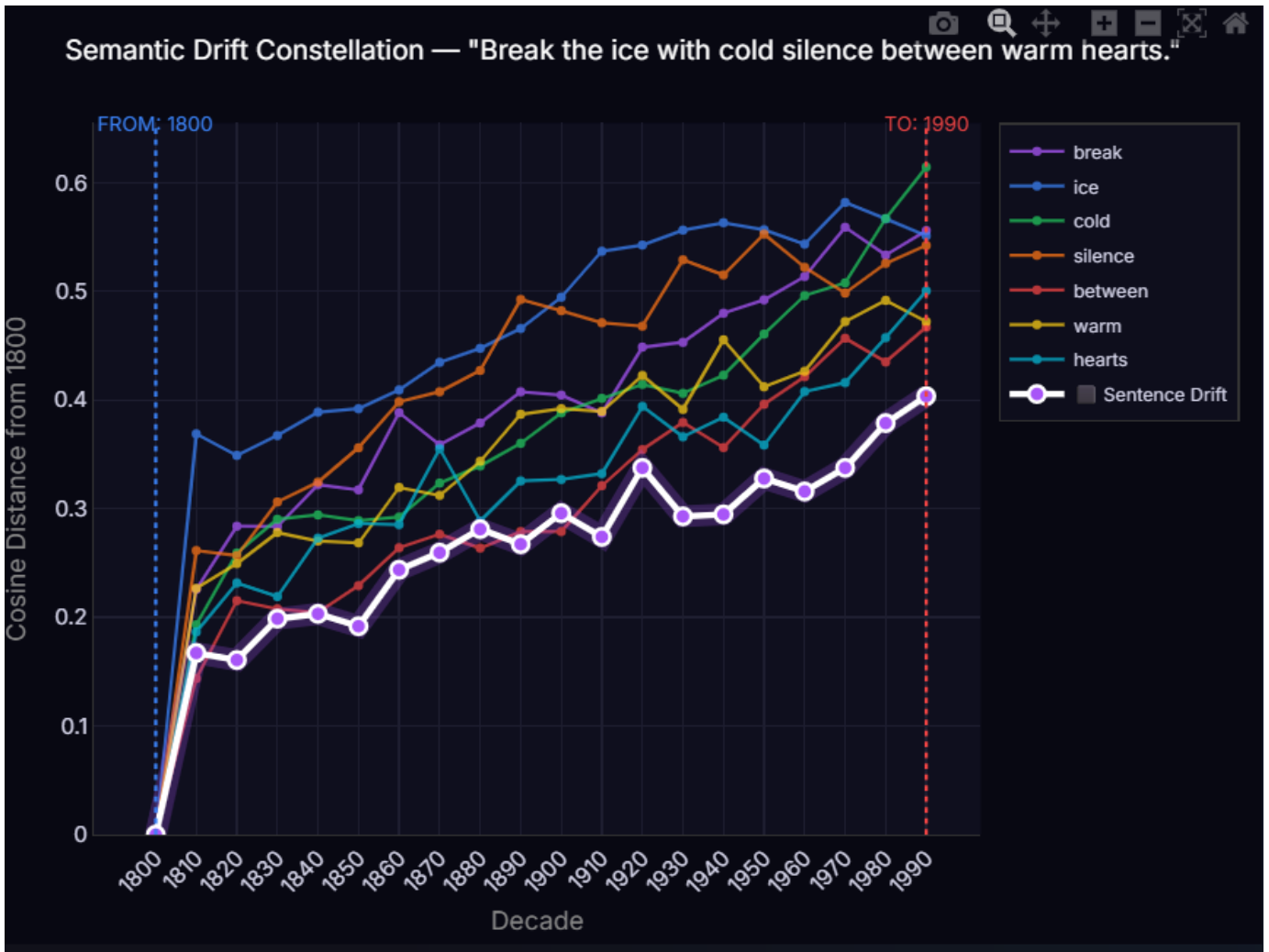


Fig. 3. Innovation: The Drift Constellation

incorporate Contextualized Embeddings via deep transformer architectures (such as Temporal BERT formulations) to navigate intrinsic historical polysemy mapping more granularly, as well as introduce cross-lingual evaluations.

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